Green University of Bangladesh

Department of Computer Science and Engineering (CSE)
Faculty of Sciences and Engineering
Semester: (Spring, Year: 2025), B.Sc. in CSE (Day)

Lab Report: 01

Course Title: Data Mining Lab
Course Code: CSE 436
Section: 212_D2

Lab Manual 01 & 02

Python Basics Programming.
&
Python Programming (Loop, Functions, Class and Object).

Student Details

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Lab Report Status
Marks : ……………………………………..Signature…………………………
Comments : ……………………………….Date :…………………………….

1. TITLE OF THE LAB EXPERIMENT:

- Write a Python program to find the sum of odd and even numbers from a set of numbers.
- Write a Python program to Check Triangle is Valid or Not.

2. OBJECTIVES/AIM:

- To write a Python program that calculates the sum of odd and even numbers from a given set/list.
- To check if a triangle is valid based on its three sides using Python.

3. MOTIVATION:

- The first task introduces iteration, conditionals, and list handling, and Motivate for problem-solving attitude.
- The second task focuses on applying mathematical rules in programming, enhancing problem-solving ability.

4. PROCEDURE:

Part 1: Sum of Odd and Even Numbers

- Take a list of numbers as input.
- Loop through the list.
- Separate odd and even numbers using modulus operator.
- Sum odd and even numbers separately.
- Print the results.

Part 2: Triangle Validity Check

- Take three sides of a triangle as input.
- Check if the sum of any two sides is greater than the third.
- If all three conditions are met, the triangle is valid.
- Display the result.

5. CODE:

Write a Python program to find the sum of odd and even numbers from a set of numbers.

```
numbers = input("Enter the numbers separeted by space: ");
```

```
Enter the numbers separeted by space: 2 3 4 6 7 8 10
```

In [4]:

```
num_list = list(map(int, numbers.split()));
```

In [5]:

```
def sum_odd_even(num_list):
    even_sum = sum(num for num in num_list if num % 2 == 0)
    odd_sum = sum(num for num in num_list if num % 2 != 0)
    return even_sum, odd_sum
```

In [6]:

```
even_sum, odd_sum = sum_odd_even(num_list)
```

In [7]:

```
print(f"Sum of Even Numbers: {even_sum}")
print(f"Sum of Odd Numbers: {odd_sum}")
```

Write a Python program to Check Triangle is Valid or Not

In [11]:

```
side1 = float(input("Enter the value side 1: "))
side2 = float(input("Enter the value side 2: "))
side3 = float(input("Enter the value side 3: "))
```

```
Enter the value side 1: 4
Enter the value side 2: 5
Enter the value side 3: 3
```

In [12]:

```
def is_valid_triangle(s1, s2, s3):
    return s1 + s2 > s3 and s1 + s3 > s2 and s2 + s3 > s1
```

In [13]:

```
if is_valid_triangle(side1, side2, side3):
    print("The triangle is valid.")
else:
    print("The triangle is NOT valid.")
```

6. Output:

```
print(f"Sum of Even Numbers: {even_sum}")
print(f"Sum of Odd Numbers: {odd_sum}")
```

```
Sum of Even Numbers: 30
Sum of Odd Numbers: 10
```

:

```
if is_valid_triangle(side1, side2, side3):
    print("The triangle is valid.")
else:
    print("The triangle is NOT valid.")
```

```
The triangle is valid.
```

7. Discussion:

In part 1, the loop iterates through a list and checks for even or odd numbers using the modulus operator `%`. The even and odd sums are maintained in separate variables and printed at the end.

In part 2, the triangle inequality theorem is applied. A triangle is valid if the sum of any two sides is greater than the third side.